Barriers between Avian & Human IAV

1. Entry

2. Immune evasion

3. Replication

這一系列的簡報是想跟大家說明病毒要跨著種感染其實有很多限制。很久以前學的時候只談entry，但其實整個viral life cycle的許多環節都可以作為host restriction的所在。
接下來幾份簡報想跟大家談的是：豬能作為mixing vessel，是因為其對avian與human IAV的接受度相較其他物種來得高，所以在豬隻身上，不同IAV相遇→重組的機會增加。那我們會分別從entry、immune evasion跟病毒複製的三個層面舉例說明。要提醒這邊只有挑幾個重要的舉例給大家，不是已知的全部，如果大家有興趣google還可以看到別的！

The mixing vessel theory

Receptor–types and distribution

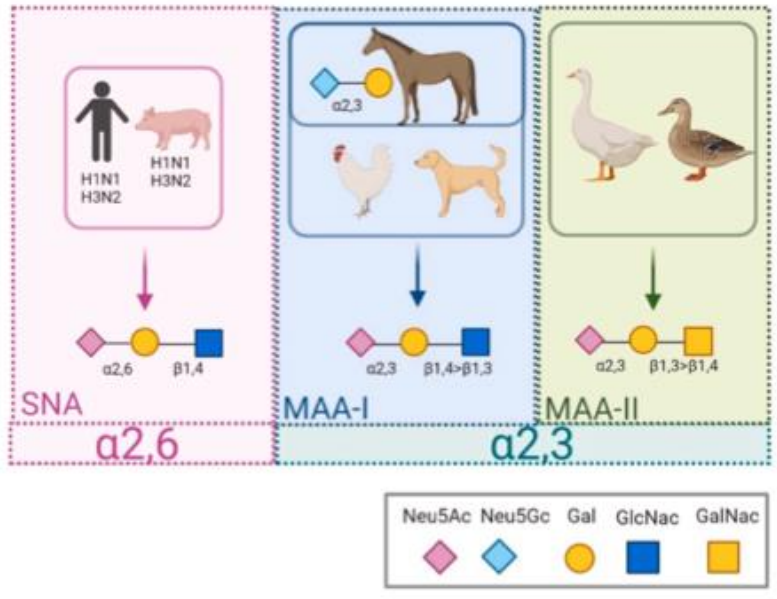


Fig. 1. Illustration of the general receptor binding preferences of different host species of influenza A virus and the main binding affinity of Sambucus Nigra Lectin (SNA), Maackia amurensis Lectin I (MAA-I), and Maackia amurensis Lectin II (MAA-II) lectins. Neu5Ac= N-Acetylneuraminic acid. Neu5Gc= N-Glycolylneuraminic acid. Gal= galactose. GlcNac= N-acetylglucosamine. GalNac= N-acetylgalactosamine. Made in Biorender.com.

Kristensen et al., 2024

IAV receptor: sialic acid

-- Avian IAV HA: α2, 3-linked

-- Human IAV HA: α2, 6-linked

-- Swine IAV HA: α2, 6-linked

-- Two critical aa mutations (190D & 225D) in HA are required for avian-to-human HA switch.

But **sufficient**?

這張簡報想說的東西其實很簡單，就是不同型別的IAV使用的receptor (sialic acid不同)
圖中可以看到human & swine IAV喜歡用的是α2, 6-linked，avian IAV則是 α2, 3-linked

根據文獻，如果avian IAV要感染人的話，receptor-binding domain上190D & 225D的兩個突變會大幅度幫助IAV與α2, 6-linked 結合

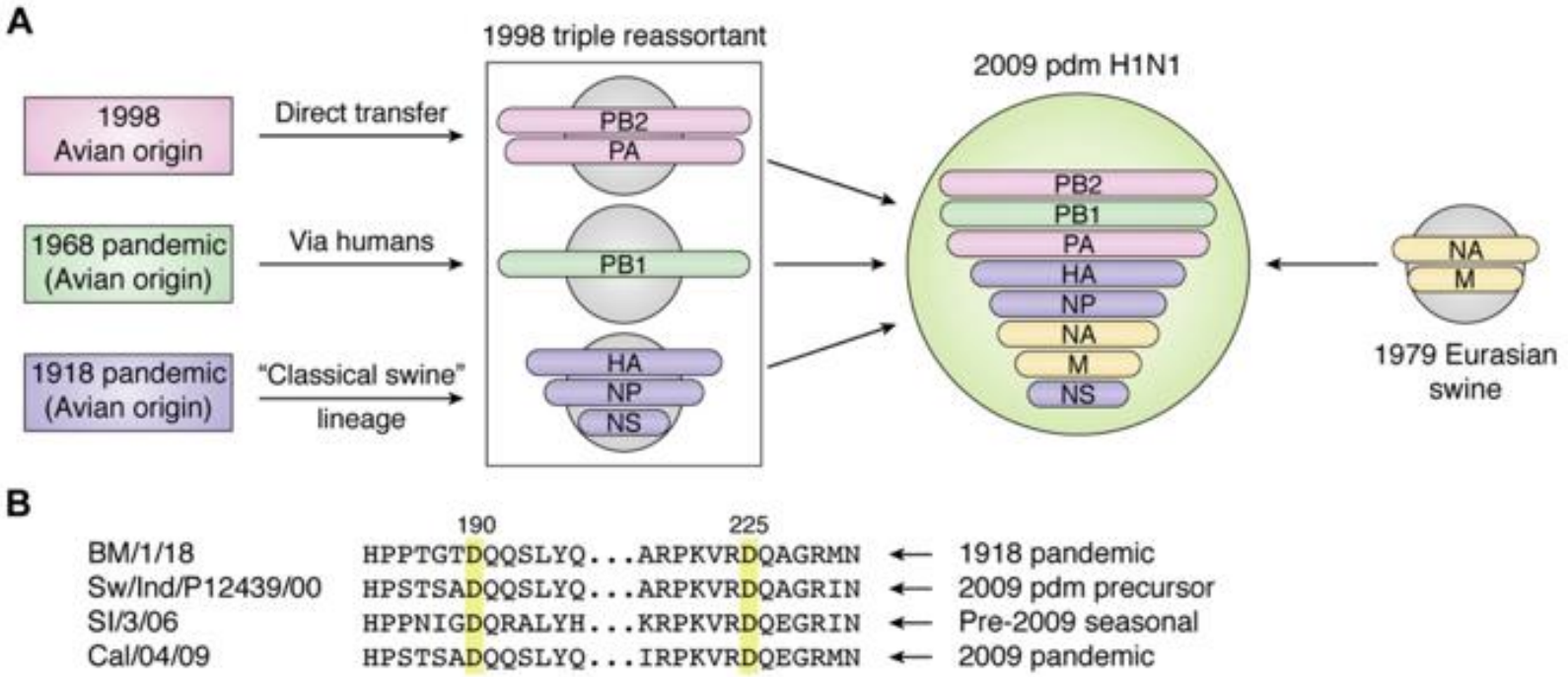


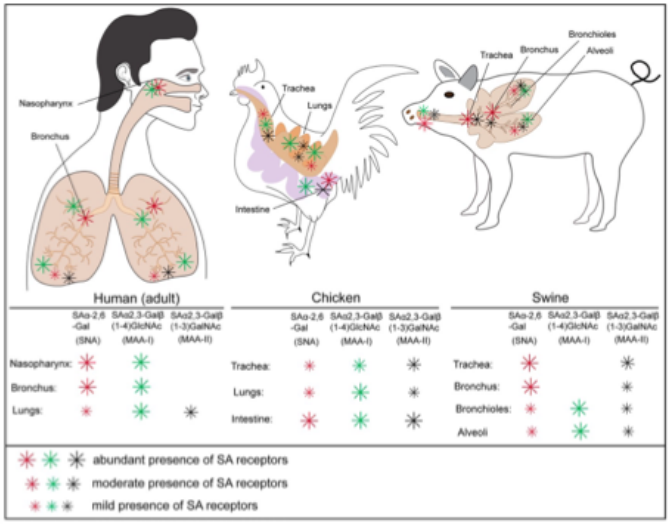
Figure 3. Reassortment leading to emergence of the 2009 pandemic H1N1 virus. *A*, cartoon schematic of various swine lineages contributing gene segments to the novel virus (adapted from data in Garten *et al.* (2009) (24)). *B*, conservation of key humanizing receptor-binding variants in example viruses from the original 1918 pandemic, through the classical swine H1N1 lineage, and within both pre- and post-2009 human H1N1 strains. Strain abbreviations are: A/Brevig Mission/1/1918, A/swine/Indiana/P12439/2000 (nearest identified swine HA gene precursor to 2009 human pandemic strains), A/Solomon Islands/3/2006 (pre-2009 seasonal), and A/California/04/2009 (2009 pandemic).

圖A將2009年pandemic swine influenza H1N1的序列（中間綠色圓形的是其不同RNA segments）進行分析，發現主要有四個來源，分別是左邊紅、綠、紫顏色背景的avian IAV以及右手邊灰色背景顏色的swine IAV。其中HA與1918西班牙流感同源，箭頭上方可以看到有註解"classical swine"，原因是豬流感被發現與後續被分離出的時空背景與1918年的西班牙流感有關。

圖B分別就幾個有意義的分離株，包括1918 pandemic、2009 pandemic前中後期的不同分離株，都可以觀察到前一頁提到，禽流感要能感染人，往往會出現的190D & 225D的突變，形成 Humanized或human-adaptive IAV。

The mixing vessel theory

Receptor–types and distribution



Modified from Chauhan1 & Gordon, 2022

-- Both human and swine respiratory tracks **contain two types of sialic acid receptors but exhibit different distribution.**

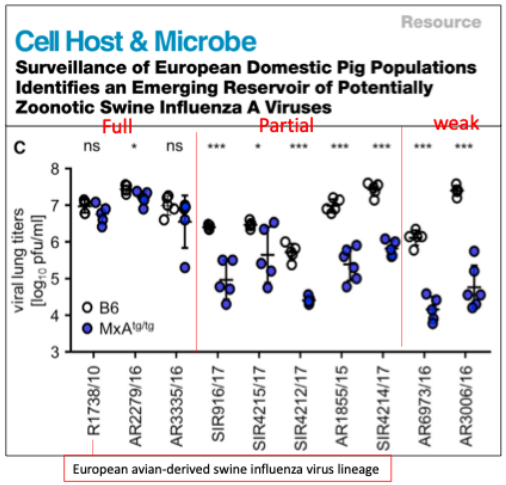
至於這張簡報想跟大家說明的是，已知IAV HA的胺基酸變化可以幫助病IAV的宿主專一性從禽類轉去感染豬或人（humanized, acquired binding from α2, 3-linked SA to α2, 6-linked SA），這是在病毒方面的轉變。那宿主的部分呢？

宿主體內專一性受體的分佈也會影響到病毒建立感染的難易性，以及病毒在宿主體內傳播的動態變化。我們常說豬同時有α2, 3-linked與α2, 6-linked SA，所以豬可以當一個mixing vessel，言下之意好像是人沒有α2, 3-linked SA，因此需要一個中間媒介。其實不然。

上圖紅色星號代表α2, 6-linked SA，而綠色跟黑色星號則都是α2, 3-linked SA，只是不同的異構物（如陸禽IAV慣用Siaα2-3Galβ1-4GlcNAc，水禽IAV慣用Siaα2-3Galβ1-3GlcNAc，但不是絕對）。可以觀察到人的呼吸道中其實還是有α2, 3-linked SA，並以1-4為主（這跟年齡其實也有關，幼童跟成人的分佈會有差）。豬的話則1-3與1-4分布相對較豐富與全面。只是如果研究已經告訴我們，人也同時有avian IAV的receptor，那單純receptor的有無便並不足以解釋為什麼豬隻可以作為mixing vessel。

The mixing vessel theory

Immune evasion

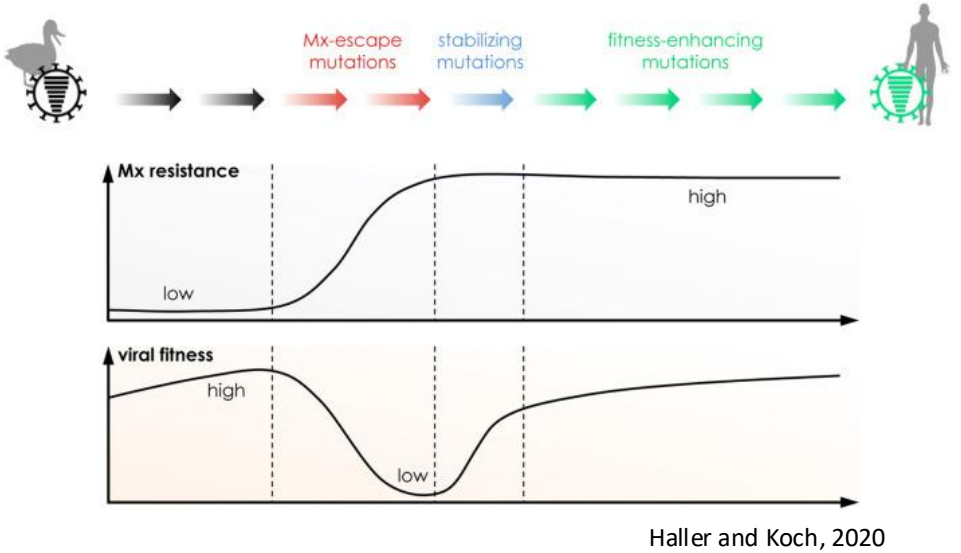


MxA = Myxovirus Resistance Protein A

-- Human MxA protein has been shown to be a key inhibitor of IAVs especially of avian origin.

-- Swine Mx1 protein shows less efficient antiviral activity against IAV comparing to human MxA.

-- Currently circulating swine IAVs could possess at least partial MxA resistance that could facilitate human infection with these viruses.



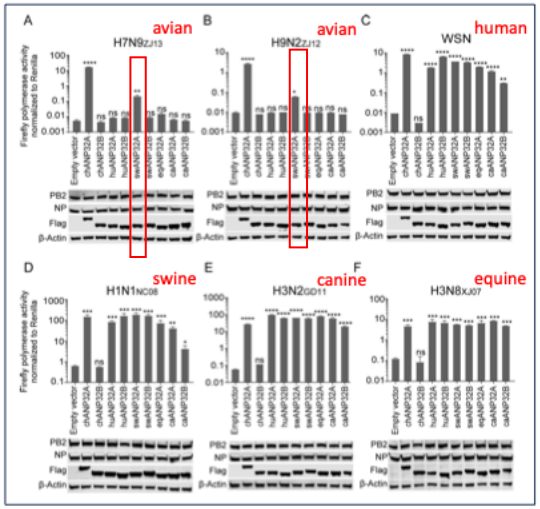
Haller and Koch, 2020

這張簡報想從免疫逃脫的部分和大家說明為什麼avian IAV比較容易在豬身上建立感染。這是一個很重要的前提，IAV的序列要發生突變需要先建立感染、複製才有機會讓polymerase出錯；要和其他IAV重組（reassort），需要不同IAV同時感染一個宿主細胞。若感染不易建立，上述前提的機會就降低。MxA是一種由interferon誘導表現的抗病毒蛋白，也是限制IAV跨物種感染的重要宿主因子之一。一般avian IAV感染人之後會受到MxA的抑制而無法有效複製，但如果今天一株avian IAV感獲得了human MxA-escape mutation，就表示病毒將有機會在人體內更有效地複製、產生子代、甚至可能造成人與人之間的有效傳播。上述假設為真，所幸事實**沒那麼簡單**，因為獲得了MxA-escape mutation常伴隨其他複製能力的減弱（右上圖紅色箭頭那欄，獲得新宿主的resistance但fitness也大幅下降），更遑論還有下一頁簡報提到複製階段的host restriction。所以除非病毒&宿主都活得夠久，進一步獲得有助fitness的second site mutation，不然可能就會僅限於單次或零星的spillover，但無法有效人傳人。

MxA作為host restriction的例子還包括，equine MxA可以抑制avian or human IAV但對equine IAV的效果就比較弱；swine MxA對avian IAV有好的抑制效果，但對swine & human IAV的效果就較差。還記得我們說reverse zoonosis其實更常發生，因為豬對human IAV的確抗性較低，這其實也反應了豬何以作為其他物種IAV跨足人類的一個基石！因為豬是累積適應性突變的溫床。左上圖是歐洲針對田間豬流感逃脫human MxA能力進行的調查，可以觀察到多數分離株都有partial resistance。

The mixing vessel theory

Replication fitness



Zhang et al, 2020

ANP32 = Acidic leucine-rich nuclear phosphoprotein 32

-- Host ANP32 protein are associated with influenza virus RNA polymerase activity and provide host restriction for mammalian species again avian influenza viruses.

-- Comparing to mammalian ANP32A&B proteins, the swine ANP32A has a stronger ability to support avian influenza virus polymerase.

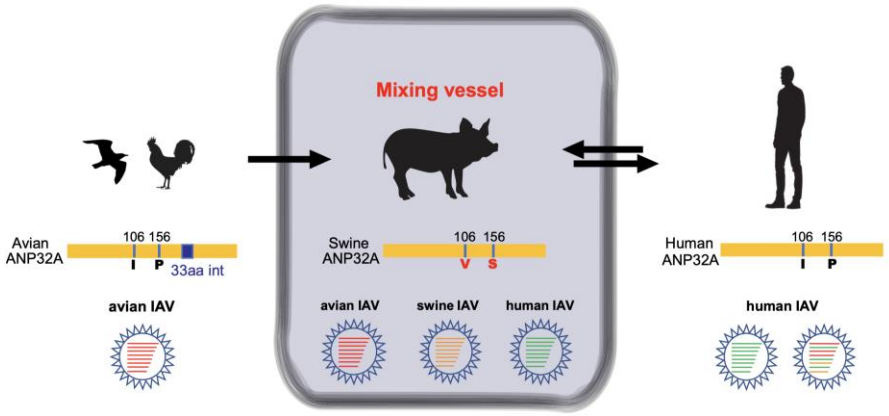


Fig 9. Proposed model of ANP32 proteins as "the key" for the inter-species transmission of avian influenza A virus in different species. AIVs can replicate and transmit naturally in birds. Pigs are thought to play an important role in influenza virus transmission and triple-reassortant virus generation, because they are susceptible to both avian and mammalian influenza viruses, and different species of influenza viruses can therefore genetically recombine in pigs, producing new strains that can transmit to humans. Avian ANP32A can support the replication of the avian influenza virus due to an additional 33-aa insertion; huANP32A only can support the polymerase activity of mammalian or mammal-adapted (e.g. PB2 E627K, or D701N) viruses, but not that of avian viruses); as swANP32A contains a unique active site 106V/156S, it can support the replication of not only all mammalian influenza viruses, but also avian influenza viruses. The co-infection of two or more virus strains in pig providing more chances to produce new recombinant strains, which can spread to other mammals such as humans.

假設一株IAV成功進入新的宿主細胞、逃脫了先天免疫的偵測，在完成生活史前他還需要面臨的大問題就是宿主細胞能否支持他完成複製跟組裝。

過去已知avian IAV在mammalian cell複製的很差，而上面圖片來源的paper主要在探討一個廣泛存在不同物種的host protein–ANP32A & ANP32B對avian IAV複製能力的影響。結果發現，相較於其他mammalian ANP32 proteins，豬源的ANP32A (not B) 可以提供avian IAV較好的複製條件（左上圖A & B，紅色框框），而這個與其他物種ANP32A的差異主要來自swANP32A上兩個胺基酸的變異（右上圖中間顯示的106V與156S）。

ANP32A是怎麼調控IAV的複製呢？ANP32A會協助IAV的polymerase 形成asymmetric dimer，幫助viral RNA的合成。不同物種ANP32A序列的差異會造成ANP32A和polymerase作用時形成defective asymmetric dimer而無法有效catalyze viral RNA的合成。

針對avian IAV，avian ANP32A相較其他mammalian ANP32A較長，來自一段33 aa的序列，與avian IAV能成功在禽類細胞中複製有關。